Unit 1: Magnetism

Summary

- A magnet always contains a pair of opposite poles, called north and south.
- Like poles repel; opposite poles attract.
- We use field lines to show the magnetic field of a magnet. The direction of the field lines is from the north pole to the south pole of the magnet.
- The closer together the field lines are, the stronger the magnetic field; field lines never cross.
- When two bar magnets are brought close together, their magnetic fields interact.
- Earth has a magnetic field that surrounds the whole planet. For the Earth, the Arctic region is the magnetic south and geographic north; the Antarctic region is the magnetic north and geographic south.
- A compass has a needle made of magnetic material that can swing around freely. The north pole of a compass always points towards the south pole of a magnet.
- The Earth's magnetosphere is the space around the Earth that is created by its magnetic field.
- The magnetosphere deflects the solar wind from the sun and protects the Earth.
- The aurora is a bright glow seen in the night skies of the polar regions.
- Magnetic storms and auroras affect space weather.

Unit 2: Electrostatics

Summary

- Uncharged objects: number of electrons = number of protons
- Negatively charged objects: number of electrons > number of protons
- Positively charged objects: number of electrons < protons
- Only electrons can be transferred during friction.
- Triboelectric charging happens when certain materials become electrically charged through friction. The object then has static electricity.
- During friction the charges are separated, but the sum of the positive and negative charges is still zero.
- Principle of Conservation of Charge: The net charge of an isolated system remains constant during any physical process.
- We use the symbol Q for electric charge and we measure electric charge in coulomb (C).
- When two small identical conducting objects on insulated stands touch, each will have the same final charge after separation: $Q = Q_1 + \frac{Q_2}{2}$
- The elementary charge is the charge on one electron = $-1{,}6 \times 10^{-19}$ C. All other charges are multiples of this elementary charge.
- Principle of Charge Quantisation: An object has an electric charge which is an integral multiple of the elementary charge: $Q = nq_e$
- Like charges repel (positive to positive or negative to negative).
- Unlike charges attract (positive to negative).
- Charged objects can attract uncharged insulators through polarisation.

Unit 3: Electric circuits

Summary

- The potential difference (V) between two points in a conductor is the work done per unit charge to move a positive charge from one point to another: $V = \frac{W}{Q}$
- The emf of a battery is the total amount of electrical energy or voltage that the battery can supply. To measure the emf we connect a voltmeter across the two terminals of the battery when *no* current is flowing.
- The terminal potential difference is the voltage measured across the terminals of a battery when current is flowing through the battery.
- The electric current (I) is a measure of the rate at which charge flows past a point or across the cross-section of a conductor: $I = \frac{Q}{\Delta t}$
- According to convention, the current flows from the positive terminal to the negative terminal of a battery.
- Voltage (potential difference) and emf are measured with a voltmeter in volts (V) connected in parallel to the battery or component it is measuring.
- Electric current is measured with an ammeter in ampere (A) connected in series to the other components in the circuit.
- Resistance is the opposition to the flow of electric current: $R = \frac{V}{I}$
- Apart from the type of material a conductor is made of, the length, thickness and temperature of a conductor affect its resistance.
- When the chemical potential energy of a battery has been depleted, the battery has gone flat.
- Resistors in series:
  Act as obstacles to the flow of charge
  - The current is the same throughout the circuit ($I_1 = I_2$)
  - The potential differences around the circuit must add up to the emf of the battery ($V_T = V_1 + V_2 + V_3$) – resistors in series are voltage dividers
  - The resistance increases as more resistors are added ($Rs = R_1 + R_2 + R_3$).
- Resistors in parallel:
  - Each resistor provides an alternative route for the current
  - The potential difference across each resistor is the same as the potential difference of the battery ($V = V_1 = V_2$)
  - The resistors split the current proportionally – the resistor with the smallest resistance will carry the largest current and the total current is the sum of all the individual currents ($I_T = I_1 + I_2 + I_3$) – resistors in parallel are current dividers
  - When two or more resistors are in parallel, the overall resistance is reduced – the total resistance will always be less than that of any individual resistor; equivalent resistance is calculated by $\frac{1}{Rp} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$; for two resistors connected in parallel: $Rp = \frac{\text{product}}{\text{sum}} = \frac{R_1R_2}{R_1 + R_2}$